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Kartu Prakerja Program and Youth Employment: Gender-Specific Outcomes in Indonesia

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Abstract

The *Kartu Prakerja* Program is a form of the active labor market policy in Indonesia, serving as a significant initiative to enhance competencies and job opportunities, including for vulnerable groups such as the young and female labor force. This study aims to analyze the program's impact on youth employment, with a focus on gender differences. Using Propensity Score Matching (PSM), the results show a significant positive effect on employment probability, particularly for women. The findings suggest that skills training and financial incentives play a crucial role in enhancing employability. The study recommends expanding program access to vulnerable groups and continuously assessing its effectiveness.

Keywords: active labour market policy, *Kartu Prakerja* Program, unemployment, youth, women

JEL classifications: J16; J24; J68

1. Introduction

The eighth Sustainable Development Goal (SDG) emphasizes promoting inclusive and sustainable economic growth, productive employment, and decent work for all. This agenda serves as a global development priority to be pursued until 2030 (United Nations 2023). In line with this agenda, Indonesia has integrated the SDG into its Medium-Term Development Plan (RPJM), reflecting the vision of the president for national development outlined in *Nawacita* (The Ministry of National Development Planning 2020). A flagship government program supporting this agenda, specifically to create productive employment opportunities, is the *Kartu Prakerja* (Pre-Employment Card) Program (APEC Economic Committee 2021).

Launched in 2020 under Active Labor Market Policies (ALMPs), the Pre-Employment Card Program was designed in response to the challenges posed

by the COVID-19 pandemic. Its primary aim is to improve workforce skills, boost productivity and competitiveness, and encourage entrepreneurship (Presidential Regulation No. 76/2020). Within two years, the program benefited more than 11 million people across Indonesia, supported by a budget of 41 trillion rupiah. The government has committed to extending this program until at least 2024, recognizing its potential to overcome ongoing labor market challenges (Coordinating Ministry for Economic Affairs of the Republic of Indonesia 2022).

This program holds particular significance as Indonesia is entering a demographic dividend stage, which is projected to peak between 2020 and 2030. During this period, the productive age population will be twice as large as the dependent population, presenting a unique opportunity to accelerate economic growth, provided these workers are efficiently absorbed into the labor market (Statistics Indonesia 2022). The absence of sufficient employment opportunities to match the large workforce may intensify existing labor market challenges (Bloom, Canning & Sevilla 2003).

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In August 2021, the unemployment rate in Indonesia stood at 6.49 percent, with 9.10 million people unemployed, while the labor force participation rate was 63.40 percent, indicating that a significant portion of the working-age population was not engaged in the labor market (Statistics Indonesia 2021). Gender disparities were notable, as only half of the working-age female population was involved in the workforce.

Youth unemployment remains another pressing issue, with 19.55 percent of young individuals unemployed in 2021. Specifically, unemployment rates for young males and females were 19.31 percent and 16.25 percent, respectively (Statistics Indonesia 2021). These figures are nearly five times higher than the adult unemployment rate, yielding a youth-to-adult unemployment ratio of 4.63, indicate challenges related to skill mismatches, lack of work experience, and limited professional networks.

The long-term consequences of these labor market barriers are substantial. Prolonged unemployment can lead to scarring effects, reducing the employability of individuals over time and increasing the risk of exclusion from the labor market (Bentolila & Jansen 2016; Nordlund & Greve 2018). Therefore, targeted interventions such as the Pre-Employment Card Program are essential to mitigate these risks and improve the long-term employment prospects of youth and women.

The existing literature on ALMPs presents mixed results across different countries. Several studies indicate that ALMPs, such as vocational training, have a positive impact on employment outcomes (Blázquez, Herrarte & Sáez 2019; Bratti et al. 2022; Burger et al. 2022; Mo Costabella 2017). These programs have been particularly effective for vulnerable groups, including young people and women (Dengler 2019a,b; Escudero et al. 2019; Pastore & Pompili 2020). However, several studies raise concerns about the effectiveness of ALMPs, especially in developing countries. McKenzie (2017) and Card, Kluve & Weber (2018) discover smaller impacts on youth, suggesting that the success of these programs depend on factors such as design,

implementation, and the broader economic context.

In Indonesia, the Pre-Employment Card Program has demonstrated promise in enhancing skills, productivity, and competitiveness of participants. However, research on its specific impact on employment outcomes for vulnerable groups such as youth and women remains limited. This study aims to fill this gap by analyzing the effect of the program on the probability of securing employment among these vulnerable groups.

Employing Propensity Score Matching (PSM), this study compares program participants with non-participants, focusing on youth and gender-specific outcomes. The analysis controls for age, gender, area of residence, education level, and job experience, providing insights into the effectiveness of the program in reducing youth unemployment and addressing gender disparities in labor market outcomes.

2. Literature Review

2.1. ALMPs and the Pre-Employment Card Program

ALMPs have long served as a central strategy for mitigating unemployment and improving labor market outcomes, specifically for vulnerable groups such as youth and women (Escudero et al. 2019). These policies aim to enhance the employability of participants through various interventions, including skills training, job search assistance, wage subsidies, and entrepreneurship support (International Labour Organization 2023). By targeting both supply and demand-side barriers to employment, ALMPs help reduce structural inequalities and provide disadvantaged individuals with better opportunities to enter or re-enter the labor market (Brown & Koettl 2015; Ernst, Merola & Reljic 2022).

The concept of active labor market intervention first gained prominence in Scandinavian countries, where governments sought to reduce unemployment through proactive labor market strategies. In

the 1990s, these strategies had expanded to other Organisation for Economic Co-operation and Development (OECD) member countries in response to rising unemployment and the associated increase in long-term unemployment (Bonoli 2017; Kluve 2016). As unemployment rates rose, the urgency to improve job prospects for the unemployed became increasingly critical. ALMPs have since become a key element of labor market policies in many countries, helping to overcome structural challenges and promote sustainable employment growth (Auer, Efendioğlu & Leschke 2005; Martin 2015).

In Indonesia, the Pre-Employment Card Program, launched in 2020 in response to the economic downturn caused by the COVID-19 pandemic, represents a major ALMP initiative designed to address the growing unemployment crisis. The program combines skills training with financial assistance, offering unemployed and underemployed individuals access to online training in various fields, such as digital literacy, entrepreneurship, and sector-specific skills. Participants also receive a stipend to support their living costs during the training period (Presidential Regulation No. 76/2020). This dual-focus approach is consistent with international best practices, where the integration of training and other relevant support has proven to produce better employment outcomes than standalone interventions (Escudero 2018; Kluve et al. 2019; Pignatti & Van Belle 2021).

The design of the Pre-Employment Card Program aligns with the theoretical foundations of ALMPs, focusing on reducing unemployment by targeting supply-side barriers (inadequate skills or education) and demand-side obstacles (hiring costs and labor demand) that hinder individuals from entering the workforce (APEC Economic Committee 2021). By offering financial incentives and skills training tailored to the needs of the evolving labor market in Indonesia, the program directly overcomes two critical barriers faced by vulnerable populations (Nguyen et al. 2023).

2.2. Prior Studies of ALMPs

Numerous studies have examined the impact of ALMPs on employment outcomes in various countries. These policies typically include interventions such as skills training, job search assistance, wage subsidies, and public employment programs. Their effectiveness depends on factors such as design, target population, and implementation.

Participation in skills training programs has a positive impact on employment outcomes (Bratti et al. 2022; Mo Costabella 2017; Destefanis, Fragetta & Ruggiero 2023; Lammers & Kok 2021). Research in several countries demonstrates that skills training significantly increases employment likelihood, specifically for vulnerable groups such as youth and women (Lindley et al. 2015; Wiśniewski 2022). Similarly, job search assistance programs, which combine training with active job placement support, have proven to improve employment prospects, especially for disadvantaged individuals (Blázquez, Herrarte & Sáez 2019; Burger et al. 2022).

On the other hand, wage subsidies and public employment programs have generated mixed results. Wage subsidies may boost short-term employment, but often fail to deliver sustainable and long-term impacts, specifically for younger and older workers (Card, Kluve & Weber 2018). Similarly, public employment programs, which offer short-term government-funded jobs, typically deliver minimal long-term benefits for participants (Dias, Ichimura & van den Berg 2013; Rotar 2021). These programs may alleviate immediate unemployment but often do not equip participants with transferable skills necessary for securing permanent positions.

In developing countries, the impact of ALMPs tend to be even more inconsistent. Skills training programs, in particular, have generally underperformed compared to expectations (McKenzie 2017). Factors such as informal labor markets, inadequate funding, and poor implementation capacity frequently hinder the effectiveness of these interventions (Escudero 2018).

Despite these challenges, ALMPs often benefit vul-

nerable groups more significantly. Studies indicate that targeted interventions can reduce youth unemployment by addressing skills mismatches and facilitating structured job search support (Al Ayyubi, Pratomo & Prasetyia 2023; Escudero et al. 2019; Focacci 2020). Women, especially those balancing caregiving responsibilities or with lower educational attainment, also benefit from ALMPs that offer flexible training options and re-entry support (Dengler 2019b; Escudero et al. 2019; Kruppe & Lang 2018).

However, the overall effectiveness of ALMPs depends heavily on program design and alignment with labor market needs (Escudero 2018). Programs that lack clear pathways to stable employment may not yield long-lasting impacts (Alegre et al. 2015; Bratti et al. 2022). This highlights the importance of sustained support beyond the initial intervention.

Research on the Pre-Employment Card Program in Indonesia is still emerging, as the program was launched during the COVID-19 pandemic. Early studies by J-PAL Southeast Asia (Alatas et al. 2021) and Presisi Indonesia (2022) focus primarily on the impact of the program on skills development, productivity, and entrepreneurship. While these studies suggest that the program has positively improved skills and competitiveness of participants, evidence regarding its specific impact on employment outcomes for vulnerable groups, such as youth and women, remains limited. Given the mixed success of ALMPs in various contexts, further research is essential to understand the effectiveness of the program in improving employment outcomes for these groups in Indonesia and its potential to mitigate the long-term effects of unemployment on the most vulnerable populations.

3. Method

This study aims to evaluate the impact of the Pre-Employment Card Program on employment outcomes among young individuals aged 18–24 in Indonesia, with a particular emphasis on the dif-

ferentiated effects by gender. To address potential issues related to selection bias, this study employed PSM, a robust method widely used in observational studies to estimate causal effects.

The sample consists of individuals eligible for the Pre-Employment Card Program, defined as those who are part of the workforce, aged 18 or older, and not currently enrolled in formal education. The study focuses specifically on individuals who completed the training component of the program, ensuring that the outcomes reflect the impact of actual participation rather than mere registration. This approach mitigates concerns regarding potential proxy test-taking practices and ensures that program completion corresponds to the intended skill development and participation.

Data were sourced from the National Labor Force Survey (SAKERNAS) conducted by Statistics Indonesia in August 2021. The year was selected because it marks the full transition of the Pre-Employment Card Program from a pandemic response to a broader labor market intervention. By 2021, the program had matured, with a large number of participants having completed their training, rendering it an ideal period for assessing its impact on employment outcomes. Furthermore, the 2021 data reflect the performance of the program during the economic recovery of Indonesia from the worst effects of the COVID-19 pandemic. This offers a more precise understanding of the role of the program in facilitating entry into a fluctuating labor market.

Of the 777,982 working-age individuals in the dataset, 4,562 respondents met the sample criteria, ensuring sufficient representation for the analysis. The observations consist of 488 male participants and 1,825 male non-participants, as well as 464 female participants and 1,785 female non-participants. This gender-based breakdown allows for a more detailed analysis of the different impacts of the program on men and women.

This study utilized three types of variables: treatment, outcome, and covariate. Program participa-

Table 1. Distribution of Observations

Observation	Number of Observations
Male observations	
Program Participants	488
Non-Program Participants	1,825
Female observations	
Program Participants	464
Non-Program Participants	1,785
Total	4,562

Source: Statistics Indonesia (2021)

tion (i.e., completion of the Pre-Employment Card Program) serves as the treatment variable, while the outcome variable measures the probability of securing employment. Covariates include key individual characteristics, such as age, gender, area classification (urban/rural), education level, and job experience.

To estimate the causal effect of the program on employment outcomes, PSM was employed to create a comparable control group of non-participants. PSM is a technique widely used in observational studies to address selection bias by approximating a randomized experiment (Heckman, Ichimura & Todd 1997; Rosenbaum & Rubin 1983). It creates a statistical counterfactual by matching treated and untreated individuals with similar observable characteristics, thereby reducing the influence of confounders on the estimated treatment effects (Caliendo & Kopeinig 2008; Heckman et al. 1998). The application of PSM in this study aims to verify that differences in employment outcomes are attributable to program participation, rather than pre-existing disparities between treated and untreated individuals.

To implement PSM, individuals were divided into a treatment group (participants in the Pre-Employment Card Program) and a control group (non-participants). Matching was performed based on a set of observable characteristics: age, gender, area classification, education level, and job experience. This process ensures similar covariate distribution between treated and control units, thereby isolating the impact of the program on employment probability. The average treatment effect

on the treated (ATT) is expressed as follows:

$$ATT = E(Y_1|P(X), D = 1) - E(Y_0|P(X), D = 0) \quad (1)$$

where Y denotes the employment outcome, D symbolizes program participation, and X represents the set of covariates. The ATT captures the effect of program participation on the employment outcomes of the treated group, controlling for observable covariates.

The matching procedure employed the Nearest Neighbor (NN) matching algorithm, which pairs treated individuals with control units that have the closest propensity scores. This method minimizes differences between matched pairs, thus improving the comparability between treatment and control groups (Khandker, Koolwal & Samad 2009). Additional robustness checks were conducted using radius matching and kernel matching to validate the consistency of the results. These different matching techniques test the robustness of the estimated impact of the program.

Two key assumptions underlie the validity of PSM estimates: conditional independence and common support. The conditional independence assumption implies that program participation is independent of potential outcomes, conditional on the observed covariates (Heckman, LaLonde & Smith 1999; Khandker, Koolwal & Samad 2009). Although perfect matching is unattainable in quasi-experimental settings, PSM allows the use of propensity scores as a summary measure of covariates, ensuring balance in observed characteristics between treatment and control groups. The common support assumption ensures that individuals in the treatment group have counterparts in the control group with similar propensity scores. This is crucial to avoid comparing incomparable individuals and to guarantee the estimation of the treatment effect for individuals who are reasonably eligible for the treatment.

The propensity score was estimated using a weighted probit model, where the dependent variable is program participation and the independent variables are the covariates (age, gender, area

classification, education, and job experience). To account for the complex sampling design of the SAKERNAS dataset, a weight variable was applied. By incorporating this weight, the propensity score estimation reflects population-level characteristics and adjusts for potential sampling bias. The propensity score equation is specified as follows:

$$Pr(\text{treatment} = 1|X) = \beta_0 + \beta_i X_{ij} + \varepsilon_j \quad (2)$$

where X_{ij} represents the covariates, β_0 is the intercept, β_i are the covariate coefficients, and ε_j is the error term.

The common support condition is validated by examining the overlap in the distribution of propensity scores between the treatment and control groups. Adequate overlap confirms that individuals in the treatment group have comparable matches in the control group, thus satisfying the common support assumption (Khandker, Koolwal & Samad 2009). To assess the quality of the matching, a balancing test was conducted post-matching to confirm that covariates are similarly distributed across the treated and control groups, thus minimizing selection bias (Caliendo & Kopeinig 2008).

While PSM helps mitigate selection bias from observable covariates, it does not account for unobserved factors that may influence both program participation and employment outcomes. To assess the potential impact of unobserved confounders, a Rosenbaum Bounds sensitivity analysis was conducted. This analysis tests the robustness of the estimated effect by varying the parameter Γ , which quantifies the potential influence of hidden bias.

In this analysis, Γ values ranging from 1 to 3 were considered. A value of $\Gamma = 1$ assumes no hidden bias, meaning that individuals with identical observed covariates have equal probabilities of receiving the treatment. Higher Γ values reflect a more pronounced influence of unobserved confounders. The objective of this analysis is to determine whether the treatment effect remains significant even when allowing for moderate unobserved biases, thus providing a measure of the robustness

of the results against potential hidden variables.

4. Results and Analysis

4.1. Descriptive Statistics

This section presents an overview of the descriptive statistics for the sample used in this study. The sample includes 952 individuals who completed the Pre-Employment Card Program (treatment group) and 3,610 individuals who did not participate (control group). Regarding gender, there are 488 male participants and 1,825 male non-participants, as well as 464 female participants and 1,785 female non-participants. This distribution ensures adequate representation of both genders in the treatment and control groups for analyzing gender-based program outcomes.

The key characteristics of the sample reveal several important features. The average age of participants in the treatment group is 21.83 years, while the control group has an average age of 21.36. This indicates that both groups have a similar age distribution, minimizing age-related bias in the analysis. The gender distribution is relatively balanced, with 50.70% male and 49.30% female participants, facilitating gender-specific analysis of the effects of the program.

Geographically, 56.01% of the sample reside in urban areas, while 43.99% live in rural areas. This distribution reflects the urban-rural labor market dynamics in Indonesia, where employment opportunities are typically more concentrated in urban areas. In terms of education, most participants have completed high school, which aligns with the target population of this program, namely individuals seeking to enhance their skills to improve their job prospects. In addition, 37.09% of the total sample report prior job experience, an important factor influencing employment outcomes, as those with previous work experience are more likely to secure employment.

When examining the data by gender, notable differences are observed. Among male participants,

Table 2. Research Variables

Variables	Definition	Measurement
<i>Dependent Variables (Outcomes)</i>		
Employment probability	Probability of undertaking an economic activity to acquire or assist in acquiring income, for at least one continuous hour within the past week.	0: Unemployed 1: Employed
<i>Independent Variable (Treatment)</i>		
Program participation	Completed the training in the Pre-Employment Card Program	0: Not a program recipient 1: Program recipient
<i>Covariate Variables</i>		
Age	Age according to the most recent birthday	Years old (ratio scale)
Gender	Biological sex	0: Female 1: Male
Area Classification	Classification of area of residence	0: Rural 1: Urban
Education level	The highest level of schooling attained.	0: Had not completed primary school 1: Elementary school 2: Middle school 3: High school 4: Diploma/ bachelor/postgraduate
Job experience	Job experience prior to enrolling in the program	0: Had no job experience 1: Had job experience

Source: Statistics Indonesia (2021)

21.10% completed the program, and 48.42% were employed by August 2021. In contrast, among female participants, 20.63% completed the program, and 35.66% were employed by the same period. These findings suggest that male participants have a slightly higher employment rate following program completion, which may reflect broader gender disparities in the labor market. Women may face additional barriers, such as caregiving responsibilities or discrimination, which limit their employment opportunities despite receiving training.

The geographic distribution of participants also reveals notable gender-based differences. The proportion of male participants living in urban areas is higher (58.41%) compared to female participants (53.53%). This urban concentration among male participants may influence their employment outcomes, as urban areas generally offer more job opportunities than rural regions. In addition, 38.78% of male participants report prior work experience, compared to 35.35% of female participants, which can further contribute to differences in employment outcomes between these two groups.

These descriptive statistics illustrate that the sample encompasses a diverse range of young individuals in Indonesia, with balanced representation

across gender, geographic location, education level, and prior job experience. This balance ensures that the analysis of the Pre-Employment Card Program provides valuable insights into its impact on youth employment, with particular attention to gender differences. These descriptive statistics serve as the foundation for the PSM analysis, which estimates the impact of the program on the probability of securing employment.

4.2. The Impact of the Program on Youth Employment and Gender Differences

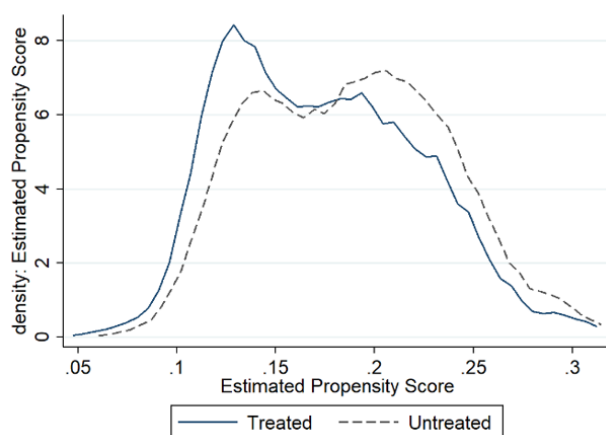
The primary objective of this study is to assess the impact of the Pre-Employment Card Program on the probability of securing employment among young participants, with a key focus on potential gender differences in the effectiveness of the program. The analysis employed the PSM method to estimate causal effects by comparing participants who completed the program (treatment group) with non-participants (control group) who share similar characteristics. Control variables include age, gender, area of residence, education level, and prior work experience. This matching technique mit-

Table 3. Descriptive Statistics

Variables	Obs	Mean	Std. Dev.	Min	Max
All observations					
Program Participation	4,562	0.2087	0.4064	0	1
Employment Status	4,562	0.3908	0.4880	0	1
Age	4,562	21.460	1.8777	18	24
Sex	4,562	0.5070	0.5000	0	1
Area Classification	4,562	0.5601	0.4964	0	1
Education Level	4,562	3.0395	0.6175	0	4
Job experience	4,562	0.3709	0.4831	0	1
Male observations					
Program Participation	2,313	0.2110	0.4081	0	1
Employment Status	2,313	0.4842	0.4241	0	1
Age	2,313	21.348	1.8493	18	24
Area Classification	2,313	0.5841	0.4930	0	1
Education Level	2,313	2.9728	0.5924	0	4
Job experience	2,313	0.3878	0.4874	0	1
Female observations					
Program Participation	2,249	0.2063	0.4047	0	1
Employment Status	2,249	0.3566	0.4791	0	1
Age	2,249	21.576	1.9000	18	24
Area Classification	2,249	0.5353	0.4989	0	1
Education Level	2,249	3.1080	0.6351	0	4
Job experience	2,249	0.3535	0.4782	0	1

Source: Calculated by the authors

igates selection bias, ensuring that any differences in employment outcomes can be attributed to the program rather than to pre-existing differences between the groups.

**Figure 1. Common Support**

Source: calculated by the authors

Propensity scores were estimated using a weighted probit model, which adjusts for the complex sampling design of the SAKERNAS dataset. Incorporating weights ensures that the estimated treatment effect is representative of the broader population,

correcting for potential sampling biases. Figure 1 illustrates the distribution of propensity scores for both the treatment and control groups, with substantial overlap between the two groups. This common support condition suggests that treated individuals have comparable counterparts in the control group, rendering the matched sample suitable for reliable estimation of the treatment effect.

Table 4. Number of Observations According to Treatment Assignment

Assignment	Off Support	On Support
All observations		
Control	0	3,61
Treated	0	952
Male observations		
Control	0	1,825
Treated	0	488
Female observations		
Control	0	1,785
Treated	0	464

Source: Calculated by the authors

Table 4 outlines the number of observations within the common support region, where 952 individuals were assigned to the treatment group and 3,610 to the control group. Disaggregated by gender, the data include 488 treated males and 1,825 male

controls, as well as 464 treated females and 1,785 female controls. This gender-based division allows for a more in-depth analysis of the differential impact of the program on men and women, which is crucial given the known disparities in labor market outcomes between genders.

The balancing tests, presented in Table 5, confirm the success of the matching process. After matching, the t-tests indicate no statistically significant differences in covariates between the treatment and control groups, demonstrating that selection bias has been substantially reduced. The average bias across all observations dropped from 12.7 percent before matching to 1.7 percent after matching. This reduction confirms that the observed differences in employment outcomes are attributable to the program rather than to pre-existing differences between participants and non-participants.

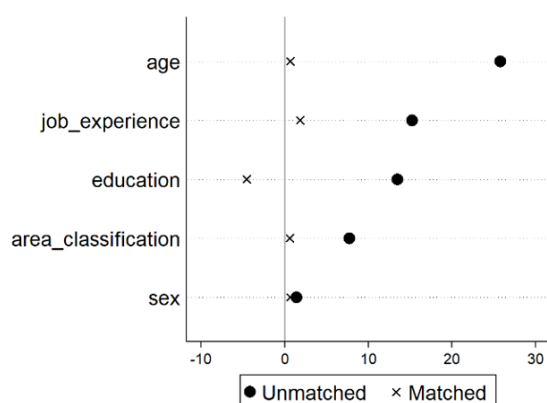


Figure 2. Selection Bias Reduction

Source: calculated by the authors

The ATT results indicate that the Pre-Employment Card Program has a significant positive impact on youth employment, increasing the probability of employment by 5.1 to 5.3 percentage points for the overall sample. This finding aligns with previous research on ALMPs, corroborating that skills training and financial support enhance employability, specifically for vulnerable groups such as youth (Card, Kluve & Weber 2018; Kluve et al. 2019). The emphasis of the program on relevant skills training, especially in digital literacy, appears to contribute

to its effectiveness. SAKERNAS data suggest that over 80 percent of the participants to increase their technical skills, likely due to the emphasis of the program on digital skills and marketable competencies (Statistics Indonesia 2021).

The analysis also reveals notable gender differences in the impact of the program. Male participants experience a 4.1 to 4.6 percentage point increase in employment probability, while female participants see a notably higher impact, ranging from 5.7 to 7.2 percentage points, with statistically significant results at the 5 percent level across all matching methods. This finding is consistent with existing research on ALMPs, which frequently demonstrates that structured training programs yield stronger benefits for women who often face greater barriers to entering the labor market (Atanassov & Trifonova 2020; Irandoust 2023; Lechner, Miquel & Wunsch 2011; Pastore & Pompili 2020).

Several factors may explain why women benefit more from the program than men. Women, particularly those in the younger age bracket, generally have less prior work experience than men, rendering them more reliant on training programs to develop marketable skills (Bernhard & Kruppe 2012; Dengler 2019b; Kruppe & Lang 2018). The Pre-Employment Card Program with its emphasis on skills development and digital literacy may have equipped women with the necessary competencies to overcome traditional barriers to workforce entry, specifically in sectors where they are underrepresented. The shift toward digital and remote work, accelerated by the COVID-19 pandemic, has further opened new opportunities for women, especially in sectors such as e-commerce and online services (Presisi Indonesia 2022).

In addition, the financial component of the program may have had a more substantial effect on women, who are more likely to face financial constraints. By reducing financial barriers to participation, the Pre-Employment Card Program enables more women to fully participate in the training and subsequent job-seeking efforts. This finding aligns with studies on similar programs, revealing that financial sup-

Table 5. t-test Results

Variable	Sample	Mean			t-test	
		Treated	Control	%bias	t	p > t
All observations						
Age	Unmatched	21.834	21.362	25.8	6.93	0.000
	Matched	21.834	21.821	0.7	0.16	0.874
Gender	Unmatched	0.5126	0.5055	1.4	0.39	0.698
	Matched	0.5126	0.5093	0.7	0.14	0.886
Area Classification	Unmatched	0.5903	0.5520	7.7	2.12	0.034
	Matched	0.5903	0.5871	0.6	0.14	0.888
Education level	Unmatched	3.1071	3.0227	13.5	3.77	0.000
	Matched	3.1071	3.1355	-4.5	-1.01	0.310
Job experience	Unmatched	0.4296	0.4204	15.2	4.22	0.000
	Matched	0.4296	0.4206	1.9	0.40	0.690
Male observations						
Age	Unmatched	21.660	21.265	21.7	4.20	0.000
	Matched	21.660	21.653	0.4	0.06	0.954
Area Classification	Unmatched	0.5943	0.5814	2.6	0.51	0.608
	Matched	0.5943	0.5975	-0.6	-0.10	0.919
Education level	Unmatched	3.0287	2.9578	11.7	2.35	0.019
	Matched	3.0287	3.0487	-3.3	-0.52	0.603
Job experience	Unmatched	0.4467	0.3721	15.2	3.01	0.003
	Matched	0.4467	0.4357	2.2	0.35	0.729
Female observations						
Age	Unmatched	22.017	21.461	30.3	5.66	0.000
	Matched	22.017	22.040	-1.2	-0.20	0.844
Area Classification	Unmatched	0.5862	0.5221	12.9	2.47	0.014
	Matched	0.5862	0.5862	0.0	0.00	1.000
Education level	Unmatched	3.1897	3.0868	16.1	3.11	0.002
	Matched	3.1897	3.1957	-0.9	-0.14	0.885
Job experience	Unmatched	0.4116	0.3384	15.2	2.95	0.003
	Matched	0.4116	0.3973	3.0	0.44	0.657

Source: Calculated by the authors

port, combined with skills training, can significantly increase women participation in the labor market (Escudero et al. 2019).

The success of the Pre-Employment Card Program for young participants can be attributed to its comprehensive design, combining skills development, financial assistance, and accessibility through on-line platforms. Studies indicate that integrating a training program with other relevant support tends to have a more sustainable impact on employment outcomes, as they simultaneously overcome multiple barriers to employment (Escudero 2018; Kluve et al. 2019; Pignatti & Van Belle 2021). The digital delivery model of the program has also been a key enabler, particularly during the COVID-19 pandemic, as it allows participants to continue their training despite mobility restrictions (Nguyen et al. 2024). Digital platforms have proven effective in reaching diverse populations and improving com-

pletion rates (Coordinating Ministry for Economic Affairs of the Republic of Indonesia 2022).

The results of the Rosenbaum Bounds sensitivity analysis further reinforce the robustness of these findings. The analysis tested Γ values ranging from 1 to 3, with $\Gamma = 1$ indicating no hidden bias. Even at $\Gamma = 3$, the impact of the program on youth employment remains statistically significant, suggesting that a considerable unobserved bias will not invalidate the results. This robustness check underscores the conclusion that the Pre-Employment Card Program positively affects employment outcomes for young participants.

In summary, the Pre-Employment Card Program has effectively addressed youth unemployment in Indonesia. Its combination of competency-based training and financial support has enabled participants to acquire relevant skills, increasing their employability in a competitive labor market. The

stronger impact observed among women highlights the importance of targeted interventions to overcome gender-specific barriers to employment. Observed from a policy perspective, expanding and refining the program, with a continued focus on vulnerable groups such as women and individuals in rural areas, can further enhance its effectiveness in reducing unemployment and fostering economic inclusion.

5. Conclusion

This study provides empirical evidence on the effectiveness of the Pre-Employment Card Program in improving employment outcomes for young people in Indonesia, particularly for young women. By employing PSM to address potential selection bias, the findings indicate that the program significantly increases the probability of employment, with female participants experiencing a greater impact than their male counterparts.

Several factors contribute to the success of the program. First, the combination of skills training and financial incentives equips participants with necessary resources to boost their employability. By prioritizing the development of marketable skills, particularly in digital literacy and technical competencies, the program helps participants to be better aligned with labor market demands. The financial support enables participants, especially those from disadvantaged backgrounds, to fully engage in training and job-seeking activities without the immediate burden of economic pressures. In addition, the digital delivery model has proven to be an effective way to reach a wide audience, particularly during the COVID-19 pandemic, when mobility restrictions hindered traditional forms of training.

Based on these findings, several recommendations are proposed to enhance the effectiveness of the program. First, efforts should prioritize expanding access for vulnerable groups, including individuals in rural areas and those with lower levels of education. While the program has proven to be particu-

larly beneficial for women, expanding its reach and tailoring content to other marginalized groups can help reduce employment inequalities. For instance, designing training programs that overcome specific labor market challenges faced by rural populations will improve both the inclusivity and effectiveness of the program.

Second, given the substantial impact of the program on the employment of women, future iterations should emphasize gender-sensitive training that targets the unique barriers women face in the labor market. Offering flexible training options, such as part-time courses or online learning, will benefit women, especially those facing discrimination in specific sectors. Customized interventions accommodating the particular needs of women can further enhance their employment outcomes.

Moreover, the program should continue to support digital and remote work opportunities, given the shift towards these types of employment due to the pandemic. The emphasis of the program on digital skills has already accelerated growth in sectors such as e-commerce and online services, which provide flexible working conditions. Expanding the emphasis of the program on these areas will further enhance its relevance and impact in the evolving labor market.

Finally, continuous monitoring and evaluation of program outcomes are essential due to the complexity of labor market dynamics. Policymakers should conduct regular assessments to identify areas for improvement and ensure that training content remains aligned with evolving demands of the labor market.

However, it is important to acknowledge the limitations of this study. As the analysis relies on cross-sectional data, PSM was employed to mitigate selection bias by matching treated and untreated individuals based on observable characteristics. While this method is appropriate for estimating program impacts, it cannot fully account for pre-treatment trends or unobserved factors that may also influence labor market outcomes. As a result, the con-

Table 6. PSM Analysis Results: The Impact of the Program on Youth Employment Probability

Matching Method	Mean of Matched Treated	Mean of Matched Controls	ATT	t-stat
All observations				
NN (5) ties	0.4422	0.3895	0.0527	2.82*
Radius (0.01)	0.4422	0.3911	0.0511	2.81*
Kernel	0.4422	0.3888	0.0534	2.94*
Male observations				
NN (5) ties	0.4693	0.4239	0.0454	1.73*
Radius (0.01)	0.4693	0.4278	0.0415	1.62
Kernel	0.4693	0.4234	0.0458	1.80*
Female observations				
NN (5) ties	0.4138	0.3417	0.0721	2.07**
Radius (0.01)	0.4138	0.3567	0.0571	2.21**
Kernel	0.4138	0.3526	0.0612	2.38**

Source: Calculated by the authors

Note: **statistical significance at 5%, *statistical significance at 10%

clusions drawn from this analysis should be interpreted with caution, especially regarding causality.

Future research can complement these findings by utilizing longitudinal data or alternative methods such as Difference-in-Differences (DiD) or Randomized Control Trials (RCTs), which allow for a more dynamic assessment of long-term effects of the program and better control for pre-existing differences between participants and non-participants. In addition, exploring the impact of the Pre-Employment Card Program across different demographic groups and regions can provide deeper insights into the distribution of its benefits.

Despite its limitations, this study provides valuable initial insights into the effectiveness of the Pre-Employment Card Program in addressing unemployment among youth and women. The results suggest that well-designed labor market interventions can help lower employment barriers and improve economic opportunities for vulnerable populations. This is particularly relevant in the wake of economic disruptions such as the COVID-19 pandemic.

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